

Calculations for Greywater Treatment (Part 2)

Important core values and assumptions:

For 1 Household / 4 People in Low-Income Indian Context

- Population Served: 4 people (1 household)
- Daily Greywater Flow (Q): 144 Liters/day (4 people x 36 Lpcd)
- Influent Phosphorus (Pin): 0.330 mg/L (Total P) (Based on average untreated greywater characteristics in India - Source 3.4)
- Influent Nitrogen (N-in): 18.98 mg/L (Total N) (Based on average untreated greywater characteristics in India - Source 3.4)
- Target Nutrient Removal Efficiency: 95% (for both N and P)
- Hydraulic Retention Time (HRT): 6.5 days (Design choice for TN removal)
- Phosphorus Uptake Rate (*Chlorella vulgaris*): 45.16 mg P L⁻¹ d⁻¹
- Algae Carbon Content (dry biomass): ~50%
- Nitrogen (N) Content in Dry Algae Biomass: 8%
- Phosphorus (P) Content in Dry Algae Biomass: 1%
- specific values for *Chlorella vulgaris* (the Phosphorus Uptake Rate, Algae Carbon Content, and the N/P content in dry biomass) were adopted from standard, commonly cited parameters found in microalgae research literature.
- These are average values considered in conceptual and mass balance calculations for *Chlorella vulgaris*
- These values can slightly change depending on cultivation conditions and parameters but are widely accepted as reliable starting points for a preliminary engineering design.

Calculations for Previous Hybrid Algal Bioreactor System Design

IMPORTANT NOTE: These calculations pertain specifically to the previous conceptual design of the Hybrid Algal Bioreactor System, which featured separate Flat Panel Photobioreactors and a distinct Airlift Column.

For the refined design, new calculations for bioreactor volume distribution, component dimensions, and pump sizing are required, as the physical configuration and integration are fundamentally different. These calculations are provided here for context and to show the evolution of the design process.

Calculations:

Phase 1: Bioreactor Volume Calculations

1. Calculate Bioreactor Volume (V) based on Hydraulic Retention Time (HRT): This volume ensures the greywater stays in the bioreactor for the desired treatment period (HRT).

Formula: $V_{HRT} = \text{Daily Greywater Flow (Q)} \times \text{HRT}$

Calculation: $V_{HRT} = 144 \text{ L/day} \times 6.5 \text{ days}$ $V_{HRT} = 936 \text{ Liters}$

2. Calculate Bioreactor Volume (V) based on Phosphorus Uptake (VP): This volume determines the minimum size needed for the algae to remove the target amount of phosphorus daily.

Step 2a: Calculate Daily Mass of Phosphorus to be Removed:

Formula: $P \text{ to remove} = P_{in} \times Q \times \text{Target Removal Efficiency}$

Calculation: $P \text{ to remove} = 0.330 \text{ mg/L} \times 144 \text{ L/day} \times 0.95$ $P \text{ to remove} = 45.144 \text{ mg P/day}$

Step 2b: Calculate Volume based on P Uptake Rate:

Formula: $VP = \text{Phosphorus Uptake Rate } P \text{ to remove}$

Calculation: $VP = 45.16 \text{ mg P L}^{-1} \text{ d}^{-1} - 45.144 \text{ mg P/day}$

$VP = 0.9996 \text{ Liters} \approx 1.0 \text{ Liters}$

3. Determine Final Target Bioreactor Volume: Compare V_{HRT} and VP . The bioreactor must be large enough to accommodate the larger of the two requirements.

Comparison: 936 Liters (from HRT) vs. ~ 1.0 Liters (from P uptake)

Decision: The HRT-based volume is significantly larger.

Final Target Bioreactor Volume: 936 Liters

Phase 2: Nutrient Removal & Biomass Production Calculations

4. Calculate Daily Mass of Nitrogen (N) to be Removed: This is the total amount of nitrogen that needs to be assimilated by the algae daily.

Formula: $N \text{ to remove} = N_{in} \times Q \times \text{Target Removal Efficiency}$

Calculation: $N \text{ to remove} = 18.98 \text{ mg/L} \times 144 \text{ L/day} \times 0.95$ $N \text{ to remove} = 2594.16 \text{ mg N/day}$

5. Calculate Daily Dry Biomass Required based on Phosphorus Removal (Biomass-P):

This calculates how much dry algae biomass needs to be produced to remove the target amount of phosphorus.

Formula: Biomass-P = P Content in Dry Algae Biomass P to remove

Calculation: Biomass-P (mg) = 0.0145.144 mg P/day

Biomass-P (mg) = 4514.4 mg dry algae/day *Convert to kg:* Biomass-P (kg) = 1,000,000 mg/kg
4514.4 mg = 0.00451 kg dry algae/day

6. Calculate Daily Dry Biomass Required Based on Nitrogen (N) Removal (Biomass-N):

This calculates how much dry algae biomass needs to be produced to remove the target amount of nitrogen.

Formula: Biomass-N = N Content in Dry Algae Biomass N to remove

Calculation: Biomass-N (mg) = 0.082594.16 mg N/day

Biomass-N (mg) = 32427 mg dry algae/day *Convert to kg:* Biomass-N (kg) = 1,000,000 mg/kg
32427 mg = 0.03243 kg dry algae/day

7. Determine Final Expected Daily Dry *Chlorella vulgaris* Biomass Production: The system must produce enough biomass to remove *both* nutrients. Therefore, we take the larger of the two biomass requirements.

Comparison: 0.00451 kg/day (from P removal) vs. 0.03243 kg/day (from N removal)

Decision: The nitrogen removal dictates the biomass production.

Final Daily Biomass Production: 0.03243 kg dry algae/day

Phase 3: CO₂ Capture Calculation

8. Calculate Mass of Carbon Contained in Algae Biomass: This determines how much carbon is locked away in the produced algae biomass.

Formula: Mass of Carbon = Final Daily Biomass Production × Algae Carbon Content

Calculation: Mass of Carbon = 0.03243 kg dry algae/day × 0.50
Mass of Carbon = 0.016215 kg C/day

9. Calculate Mass of CO₂ Captured: This converts the mass of carbon in the algae to the equivalent mass of CO₂ captured from the atmosphere (using molecular weights: C=12, O=16, so CO₂=44).

Formula: Mass of CO₂ captured=Mass of Carbon×Molecular Weight of C Molecular Weight of CO₂

Calculation: Mass of CO₂ captured=0.016215 kg C/day×1244 Mass of CO₂ captured=0.059455 kg CO₂ /day

Summary of Final Calculated Outputs:

- **Final Target Bioreactor Volume: 936 Liters**
- **Final Daily Dry *Chlorella vulgaris* Biomass Production: 0.03243 kg/day**
- **Daily CO₂ Capture Potential: 0.059455 kg CO₂/day**